



CODEALPHA CYBER SECURITY INTERNSHIP | TASK 2

Phishing Awareness Training

Understanding phishing attacks, social engineering, and how to stay safe online.

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Project Overview

Why this matters

Phishing is one of the most common cyber attacks used to steal sensitive information such as passwords, banking details, and personal data.

This presentation explains how phishing works, how to identify phishing attempts, and how to protect yourself from becoming a victim.

91%

of cyberattacks start
with a phishing email

#1

most common social
engineering attack vector

Topics Covered

An end-to-end look at phishing — from theory to real-world defense

1

Introduction to Phishing

2

Types of Phishing Attacks

3

How to Identify Phishing Emails

4

Fake Websites & URL Analysis

5

Social Engineering Techniques

6

Real-World Phishing Examples

7

Best Practices to Prevent Phishing

8

What to Do If You Become a Victim

9

Interactive Quiz

Introduction to Phishing



Phishing is a form of cyber attack where criminals impersonate trusted entities to trick victims into revealing sensitive information or installing malware.



Deceptive Contact

Attackers pose as banks, companies, colleagues, or government agencies.



Urgency & Fear

Messages create pressure to act fast — 'your account will be suspended'.



Data Theft

The goal is usually credentials, card numbers, or malware installation.

Types of Phishing Attacks

Attackers adapt the channel and target — but the goal stays the same

1

Email Phishing

Mass fraudulent emails impersonating trusted brands.

2

Spear Phishing

Highly targeted attacks aimed at a specific person.

3

Whaling

Targets executives and senior leadership.

4

Smishing

Phishing carried out via SMS text messages.

5

Vishing

Voice-call scams impersonating banks or support.

6

Clone Phishing

A legitimate email is copied and re-sent with malicious links.

How to Identify Phishing Emails

Red flags that should make you pause before you click



Generic greetings like 'Dear Customer' instead of your name



Sender address that looks close, but not identical, to the real one



Urgent or threatening language demanding immediate action



Unexpected attachments or links you weren't expecting



Poor grammar, spelling errors, and inconsistent formatting



Requests for passwords, OTPs, or payment information

Fake Websites & URL Analysis

Learn to read a URL before you trust the page behind it

SUSPICIOUS

`http://secure-paypa1-login.com`

LEGITIMATE

`https://www.paypal.com/login`

What to check

- Misspelled domain names or extra characters
- Missing HTTPS / padlock icon
- Unusual subdomains before the real brand name
- Suspicious top-level domains (.xyz, .top, .info)



Before you click, ask:

Does the domain exactly match the brand?

Am I expecting this link or attachment?

Does hovering reveal a different address?

Is the page asking for info it shouldn't need?

Social Engineering Techniques

Phishing exploits human psychology as much as technology

A

Authority

Impersonating a boss, bank, or government official to demand compliance.

U

Urgency

Creating a false deadline so victims act before they can think.

F

Fear

Threatening account suspension, fines, or legal action.

T

Trust

Exploiting familiarity with a known brand, colleague, or friend.

C

Curiosity

Using enticing subject lines like 'You've won a prize!'

R

Reciprocity

Offering a favor or gift to lower the victim's guard.

Real-World Phishing Examples

Notable incidents that show the real cost of phishing

1

Google & Facebook (2013–2015)

A fake vendor invoice scheme tricked employees into wiring over \$100 million to attacker-controlled accounts.

2

Twitter Bitcoin Scam (2020)

Attackers used phone-based social engineering (vishing) to gain access to internal admin tools.

3

Sony Pictures (2014)

Spear-phishing emails to employees gave attackers an entry point that led to a major data breach.

Best Practices to Prevent Phishing

Simple habits that stop most phishing attempts before they succeed



Verify sender identity before clicking links or attachments



Hover over links to preview the real destination URL



Enable multi-factor authentication on all important accounts



Keep software, browsers, and antivirus tools up to date



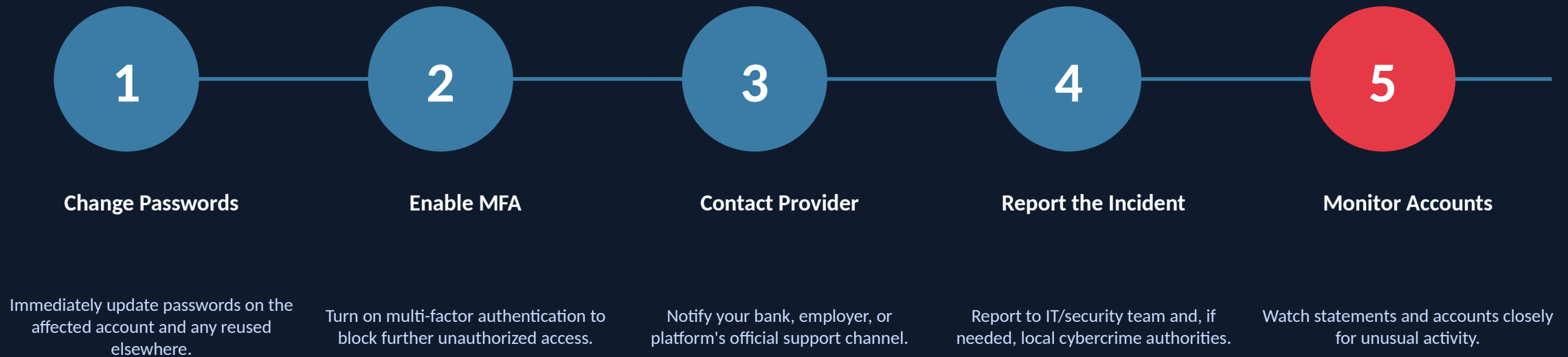
Never share passwords or OTPs over email or phone



Report suspicious emails to your IT or security team

What to Do If You Become a Victim

Fast, calm action limits the damage



Interactive Quiz

Test what you've learned — discuss answers as a group

Q1

A bank email urgently asks you to confirm your PIN by replying. What do you do?

Answer: Never reply with sensitive data — contact the bank directly using a verified number.

Q2

You receive a text saying you've won a prize and must click a link now. Is this safe?

Answer: No — urgency plus an unexpected prize is a classic smishing red flag.

Q3

A colleague's email asks for gift cards urgently, but the tone feels off. Next step?

Answer: Verify through a separate channel (call or in person) before taking any action.

Learning Outcomes

By the end of this training, participants will be able to:



Understand phishing attacks and their impact



Identify common phishing techniques



Recognize suspicious emails and fake websites



Learn effective methods to protect personal information



Improve overall cybersecurity awareness

Thank You

Stay alert. Verify before you trust. Report what looks suspicious.

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